

[A101 - Toward Modifications of Extreme Weather Events](#)

Title: Weather Jiu-Jitsu: Steering Tropical Cyclones with Small Perturbations of the Jet Stream in Experiments with Deep Learning Foundation Models

Abstract:

Deep learning foundation models provide a powerful new framework for modeling and understanding complex geophysical phenomena. By training on extensive, diverse, and multiscale Earth system datasets, these models effectively capture high-dimensional spatiotemporal dynamics without relying explicitly on physical equations. Recent advancements have shown their exceptional capabilities in global weather forecasting and accurate tropical cyclone (TC) tracking.

In this study, we investigate the application of foundation models to not only predict but actively influence the trajectories of tropical cyclones, which are among the most devastating and dynamically sensitive weather events on Earth. Our methodology applies the concept of "Weather Jiu-Jitsu," an approach that advocates using small, precisely timed, low-energy interventions to gently steer extreme weather events away from vulnerable populations and critical infrastructure. We examine historical hurricane trajectories and associated large scale mid-latitude circulation over the hurricane evolution to identify cases where there was catastrophic impact on landfall, and there was a possibility of modifying the hurricane steering winds such that the track could be controlled to avoid landfall. Using the Aurora Foundation Model we explore how nudges applied at different locations well upstream of the hurricane location could evolve to shift the steering winds so the track stays clear of landfall. Conditions under which this was potentially feasible/infeasible are identified and the prospects for practical application are discussed.

We invite collaboration to develop Weather Jiu Jitsu as a community research platform.